

WEYL GROWTH AND TRACIAL LIMITS FOR ARITHMETIC TRANSLATIONS

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ABSTRACT. We determine the growing-window tracial pressure of the actual prime-power translation operator attached to a fixed holomorphic cuspidal newform of trivial character. Every intermediate translation is killed at the interval boundary. If the interval has length ℓ , the logarithmic tracial exponential at fixed coupling s is $s^2 \ell^2 / 12 + O_{f,s}(\ell \sqrt{\log \ell})$, with a bounded lower error. The corresponding logarithmic Weil operator has Weyl coefficient $K_{f,\ell}$ satisfying $\log K_{f,\ell} = \ell^2 / 48 + \log(2\ell / \sqrt{N}) + O_f(\ell \sqrt{\log \ell})$. A uniform coupling estimate gives the spectral large-deviation law at every diverging intermediate scale up to ℓ^2 . At the central scale ℓ , the arithmetic trace converges to the $\sqrt{a}$ variance-profile Toeplitz law, whose second and fourth moments are $1/3$ and $4/15$. We determine its logarithmic tail coefficient, equal to -3 in both directions. A sharp weighted autocorrelation identity explains the pressure constant. We prove global L^2 stability modulo affine phase, with optimal square-root exponent and sharp small-deficit squared-distance coefficient 6 . An explicit taper family preserves total leading prime variance while changing the boundary pressure. Higher prime powers and ramification change the logarithmic pressure by a bounded amount, without being small in operator norm. Fixed tuples of inequivalent forms give an orthogonally invariant, noncommuting colored limit. The proofs retain boundary survival through a uniform finite-grid comparison; fixed-moment convergence alone is not used to exchange the window and coupling limits.

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1. INTRODUCTION

For an operator whose principal growth is logarithmic in frequency, a bounded perturbation can change the leading Weyl coefficient. The coefficient records an exponential statistic of the perturbation, not merely its average. When the perturbation is a sum of translations on an interval, the relevant average also remembers the boundary: a path contributes only if every one of its intermediate vertices remains inside the interval. Arithmetic translations make this geometry particularly rigid. Their lengths are logarithms of prime powers, and unique factorization determines exactly which words can return to their starting point.

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This paper studies that coefficient as the interval grows. The translation alphabet then grows exponentially, its entries have arithmetic signs, and a fixed-alphabet Weyl formula does not estimate the resulting coefficient. Our main result is a sharp pressure law for this growing arithmetic operator. A second result identifies its central tracial distribution with a known variance-profile Toeplitz law and determines the sharp logarithmic tails of that law. These are two different limiting questions. We give a uniform estimate which also fills the intermediate spectral scales.

The arithmetic operator and the main estimates. Fix a primitive holomorphic cuspidal newform f of even weight $2\kappa \geq 2$, exact level N , and trivial character. We use its real, unitary-normalized Hecke coefficients. Write α_p, β_p for its good local parameters, so $\alpha_p \beta_p = 1$, $|\alpha_p| = |\beta_p| = 1$, and $\lambda_f(p) = \alpha_p + \beta_p$. For $I_\ell = (0, \ell)$ let

$$(S_h u)(x) = \mathbf{1}_{I_\ell}(x - h)u(x - h), \quad x \in I_\ell,$$

where u is extended by zero. In particular $S_h^* = S_{-h}$. The operator under study is the finite selfadjoint sum

$$W_{f,\ell} = \sum_{p^m < e^\ell} b_f(p^m)(S_{m \log p} + S_{-m \log p}). \quad (1.1)$$

At good primes, $b_f(p^m) = (\log p)p^{-m/2}(\alpha_p^m + \beta_p^m)$; the actual ramified coefficients are specified in Section 2. These are logarithmic-derivative coefficients, not the ordinary Hecke coefficients at p^m .

The translation algebra has a normalized positive trace τ_ℓ . It is not the Hilbert-space trace. If a word has successive displacements $a_0 = 0, a_1, \dots, a_r$, its trace is

$$\mathbf{1}_{\{a_r=0\}} \left(1 - \frac{\max_j a_j - \min_j a_j}{\ell} \right)_+. \quad (1.2)$$

We construct this trace on the actual operators below. Put

$$P_{f,\ell}(s) = \log \tau_\ell(e^{sW_{f,\ell}}).$$

Our uniform estimate is the following precisely quantified assertion: for each fixed f and $C > 0$, there are constants $A_{f,C}$ and $\ell_0(f, C)$ such that, whenever $\ell \geq \ell_0(f, C)$ and $2 \leq |t| \leq C\ell$,

$$\frac{t^2}{12} - A_{f,C} \leq P_{f,\ell}(t/\ell) \leq \frac{t^2}{12} + A_{f,C}(1 + |t|\sqrt{\log(2 + |t|)}). \quad (1.3)$$

The form and C are fixed; the estimate is uniform in the displayed range of t and ℓ . There is no assertion of a threshold uniform over varying levels or weights. In particular, for every fixed real s ,

$$\lim_{\ell \rightarrow \infty} \ell^{-2} P_{f,\ell}(s) = \frac{s^2}{12}. \quad (1.4)$$

For $s \neq 0$ its lower error is bounded and its upper error is $O_{f,s}(\ell\sqrt{\log(2 + \ell)})$; $s = 0$ is exact.

The exterior-zero logarithmic Weil operator associated with this normalization has fixed-window counting function $N_{f,\ell}(E) \sim K_{f,\ell} e^{E/2}$, where

$$K_{f,\ell} = \frac{2\ell}{\sqrt{N}} \tau_\ell(e^{W_{f,\ell}/2}). \quad (1.5)$$

We recall the fixed-window input and verify its normalization in Section 2. Thus

$$\frac{\ell^2}{48} + \log \frac{2\ell}{\sqrt{N}} - O_f(1) \leq \log K_{f,\ell} \leq \frac{\ell^2}{48} + \log \frac{2\ell}{\sqrt{N}} + O_f(\ell\sqrt{\log(2 + \ell)}). \quad (1.6)$$

The high-energy limit defining $K_{f,\ell}$ is taken first, at a fixed window. The subsequent limit in (1.6) does not give a window-uniform onset for the counting asymptotic.

Central, intermediate, and exponentially weighted scales. The probability spectral laws of $W_{f,\ell}/\ell$ in τ_ℓ converge, with all moments and all real exponential moments, to one symmetric measure ν , independent of f . It is the expected limiting spectral law of the Hermitian Gaussian Toeplitz matrices

$$T_n = \sum_{j=1}^{n-1} \frac{\sqrt{j}}{n} (z_j S_j + \bar{z}_j S_j^*), \quad \mathbb{E}|z_j|^2 = 1, \quad \mathbb{E}z_j^2 = 0, \quad (1.7)$$

where the z_j are independent circular Gaussians and the shifts act on $\mathbb{C}^n$. Its first nonzero moments and sharp tail rates are

$$\int x^2 d\nu = \frac{1}{3}, \quad \int x^4 d\nu = \frac{4}{15}, \quad \lim_{R \rightarrow \infty} \frac{\log \nu([R, \infty))}{R^2} = \lim_{R \rightarrow \infty} \frac{\log \nu((-\infty, -R])}{R^2} = -3. \quad (1.8)$$

This is not a Gaussian distribution: a Gaussian of variance $1/3$ has fourth moment $1/3$.

Between this central scale and the scale ℓ^2 , choose any positive $r_\ell \rightarrow \infty$ with $r_\ell = O(\ell)$. The spectral laws of $W_{f,\ell}/(\ell r_\ell)$ satisfy a large-deviation principle with speed r_ℓ^2 and rate $3x^2$. No lower rate of divergence for r_ℓ is imposed. At the endpoint $r_\ell = \ell$ this determines the rare tracial spectral mass carrying the Weyl coefficient. For the central limiting measure, a separate positive-word comparison proves

$$\frac{s^2}{12} \leq \log \int e^{sx} d\nu(x) \leq \frac{s^2}{12} + O(|s| \sqrt{\log(2 + |s|)}) \quad (|s| \rightarrow \infty). \quad (1.9)$$

This last statement is not obtained by interchanging a fixed-moment limit with a growing coupling.

There is a useful robustness statement beyond the untapered operator. For $g_\varepsilon(a) = \sqrt{1 + \varepsilon \cos(2\pi a)}$, with $-9/20 \leq \varepsilon \leq 24/(4\pi^2 - 3)$, multiply each coefficient at p^m by $g_\varepsilon(m \log p / \ell)$. The total leading first-prime variance remains $\ell^2/2$, but the pressure constant becomes $1/12 - \varepsilon/(4\pi^2)$. Section 9 derives this from a general boundary-profile criterion. This is an explicitly defined deformation, not an alternative formula for the untapered Weil operator.

Why the boundary changes the constant. If every surviving fraction in (1.2) were replaced by one, primewise balance would give an independent-circle product with logarithm $s^2 \ell^2/2 + O_{f,s}(1)$. The boundary reduces its leading coefficient from $1/2$ to $1/12$. The reduction is determined by the weighted autocorrelation problem

$$\sup_{\|u\|_{L^2(0,1)}=1} \int_0^1 a \left| \int_0^{1-a} \overline{u(x)} u(x+a) dx \right|^2 da = \frac{1}{12}. \quad (1.10)$$

We give an exact nonnegative deficit identity and show that the maximizers are precisely the flat-modulus linear-phase profiles. More strongly, if $D(u) = 1/12 - J(u)$ for a unit profile and $d(u)$ is its L^2 distance to that family, then

$$d(u)^2 \leq CD(u), \quad d(u)^2 \leq 6D(u) + O(D(u)^{3/2}) \quad \text{as } D(u) \downarrow 0. \quad (1.11)$$

The global constant is absolute but not numerically specified; the local coefficient 6 is optimal. The proof combines weak quartic splitting modulo modulation with an explicit transverse Hessian calculation. It requires more than the weak modulus control visible in the deficit identity alone. A separate finite-dimensional deficit identity controls the variance of every vector of the Gaussian comparison matrix, not just a proposed test vector.

Two further distinctions are essential. First, powers of one prime share one phase: they are not independent random coefficients. Their complete contribution can nevertheless be restored at bounded logarithmic cost by a relative tracial Taylor estimate. Second, rounding translation lengths can create false returns. We keep an independent phase for each prime while rounding, so that these aliases are removed exactly, and enlarge the interval to dominate every survival mask multiplicatively. An additive error in each word would not suffice at exponential scale.

Relation to prior work and scope of the contribution. Laptev and Weth [13] treat the exterior-zero logarithmic Laplacian; the bounded translation-algebra extension used in (1.5) is established in [7]. Here the new problem is the growth of its actual arithmetic coefficient as the window and alphabet grow. The arithmetic interval operator and its critical heat residue also appear in [6]. Our proof uses Deligne's local bound [9] and the fixed-form Rankin–Selberg prime number theorem in the form of Gillespie and Ji [11].

Bryc, Dembo and Jiang [5] obtain the classical scalar law, and Bose, Saha and Sen [4] explicitly allow variance profiles. Their theorem contains the scalar law in (1.7) as the profile $\sigma(a) = \sqrt{a}$. Colored pairing limits and the failure of asymptotic freeness for Toeplitz matrices are already studied by Bose, Hazra and Saha [3]. We prove the occurrence of these structures in the original arithmetic translation trace, including every local factor, and give the profile-specific pressure and tail calculation.

The unweighted problem of Garsia, Rodemich and Rumsey [10] appears in the Toeplitz norm analysis of Sen and Virág [18]; Onatski [17] studies related rectangular geometries. Manov [15] treats weighted linear functionals on positive definite functions. Our quadratic functional retains both the lag weight and the square-autocorrelation constraint. The direct linear relaxation does not give its sharp constant, as explained in Section 4.

The Gaussian tools are standard concentration and matrix-series estimates [12, 14, 19]. The specific analytic work is a uniform comparison that retains killed paths, controls the actual prime variances in moving bins, and permits a growing exponential coupling. The large-deviation conclusions are consequences of that pressure estimate by exponential tilting, whose proof is included. All statements are unconditional for the fixed holomorphic forms specified above, including forms with complex multiplication.

2. THE INTERVAL TRACE AND THE WEYL COEFFICIENT

A trace on the concrete translation algebra. Let $\mathcal{A}_\ell$ be the unital norm-closed algebra on $L^2(I_\ell)$ generated by the S_h and bounded multiplication operators. Products before closure have the form $M_g S_h$ with the appropriate overlap indicators absorbed in g . The definition does not require a unique representation as a finite sum. For $v_t(x) = \ell^{-1/2} e^{itx}$ define on finite sums

$$\tau_\ell(B) = \lim_{R \rightarrow \infty} \frac{1}{2R} \int_{-R}^R \langle v_t, B v_t \rangle dt. \quad (2.1)$$

The inner product is conjugate-linear in its first argument. For a monomial the integrand is e^{-ith} times its surviving coefficient integral divided by ℓ . Thus the limit is zero if $h \neq 0$, and is $\ell^{-1} \int g$ if $h = 0$. Each approximating functional is positive, unital, and contractive. The limit therefore extends continuously to $\mathcal{A}_\ell$. For two monomials the traces of their products vanish unless the net displacement is zero; in that case a translation of the integration variable proves cyclicity. Linearity and continuity give a tracial state on the full algebra.

If $a_j = h_1 + \dots + h_j$, then $S_{h_1} \dots S_{h_r}$ acts at x only when all $x - a_j$ lie in I_ℓ . For a closed word this set of starting positions has length $(\ell - \max a_j + \min a_j)_+$. This proves (1.2). The formula keeps the boundary at each step; it is generally false that $S_h S_k = S_{h+k}$.

For a bounded selfadjoint $B \in \mathcal{A}_\ell$, the restriction of τ_ℓ to its continuous functional calculus determines a probability measure. We use that measure to interpret spectral indicators. Equivalently one may pass to the finite von Neumann algebra of the tracial GNS representation, after dividing out the trace-null ideal. Its normal trace defines $\tau_\ell(\mathbf{1}_E(B))$ for every Borel set E . No trace on all of $\mathcal{B}(L^2(I_\ell))$ is asserted.

Actual local factors. At $p \nmid N$ let the unitary Euler factor be $((1 - \alpha_p p^{-s})(1 - \beta_p p^{-s}))^{-1}$. If $p \parallel N$, it is $(1 - \epsilon_p p^{-s-1/2})^{-1}$ with $\epsilon_p \in \{1, -1\}$; for trivial character it has degree zero when

$p^2 \mid N$; these are the classical newform local factors [1]. Consequently

$$b_f(p^m) = \begin{cases} (\log p)p^{-m/2}(\alpha_p^m + \beta_p^m), & p \nmid N, \\ \epsilon_p^m(\log p)p^{-m}, & p \parallel N, \\ 0, & p^2 \mid N. \end{cases} \quad (2.2)$$

The operator (1.1) is bounded at each fixed ℓ because it is a finite sum. Terms with $p^m = e^\ell$ vanish as operators, so either endpoint convention gives the same $W_{f,\ell}$. We write $W_{f,\ell}^{(1)}$ for its good first-prime part and put

$$b_p = \lambda_f(p) \frac{\log p}{\sqrt{p}}, \quad h_p = \log p, \quad W^{(1)} = \sum_{p \nmid N, h_p < \ell} b_p(S_{h_p} + S_{-h_p}). \quad (2.3)$$

In expressions involving b_p , unqualified prime sums run over primes good for f .

The fixed-window input and its normalization. Use $\widehat{u^0}(\xi) = \int_{\mathbb{R}} e^{-ix\xi} u^0(x) dx$. The exterior-zero logarithmic operator H_{I_ℓ} is defined by the closed semibounded form with symbol $\log|\xi|$ and Fourier factor $(2\pi)^{-1}$. Its negative part is bounded because $|\widehat{u^0}(\xi)| \leq \sqrt{\ell} \|u\|_2$ and $\log|\xi|$ is integrable near zero. Its form domain is the space of supported L^2 functions with finite positive logarithmic Fourier energy. This is not the spectral logarithm of a Dirichlet Laplacian.

The fixed-window result we use from [7], in one dimension, states that for $d > 0$, $c \in \mathbb{R}$, bounded selfadjoint $B \in \mathcal{A}_\ell$, and compact selfadjoint R ,

$$N_{dH_{I_\ell} + c + B + R}(E) \sim \frac{\ell}{\pi} e^{-c/d} \tau_\ell(e^{-B/d}) e^{E/d} \quad (E \rightarrow \infty). \quad (2.4)$$

All the data in this statement are fixed while E tends to infinity. This external fixed-window input, like the arithmetic prime number theorem used below, is separated from the new growing-window estimate.

Let $\psi = \Gamma'/\Gamma$. Define $A_{f,\ell}$ on the same logarithmic form domain by the Fourier symbol

$$\log N - 2 \log(2\pi) + 2\Re\psi(\kappa + i\xi)$$

and the bounded perturbation $-W_{f,\ell}$. Since

$$r_\kappa(\xi) = 2\Re\psi(\kappa + i\xi) - 2 \log|\xi| \in L^1(\mathbb{R}),$$

its compressed multiplier $R_{\kappa,\ell}$ is trace class. Indeed, factor it as $T^* M_{\text{sgn } r_\kappa} T$, where $Tu = (2\pi)^{-1/2} |r_\kappa|^{1/2} \widehat{u^0}$; then $\|T\|_{\text{HS}}^2 = \ell \|r_\kappa\|_1 / (2\pi)$. The integrability follows from the digamma asymptotic at infinity and the integrable logarithm at zero [16]. Thus, as an identity of closed forms and their operators,

$$A_{f,\ell} = 2H_{I_\ell} + (\log N - 2 \log(2\pi))I - W_{f,\ell} + R_{\kappa,\ell}. \quad (2.5)$$

Using $d = 2$ in (2.4) gives exactly (1.5). No uniform bound on $\|R_{\kappa,\ell}\|_1$ as the window grows is needed for this fixed-window identification.

3. ARITHMETIC VARIANCE AND BOUNDED LOCAL RESTORATION

The quantitative prime input. For our fixed form the Rankin–Selberg prime number theorem gives

$$A_f(X) := \sum_{\substack{p \leq X \\ p \nmid N}} \lambda_f(p)^2 \log p = X + O_f(X e^{-c_f \sqrt{\log X}}), \quad c_f > 0. \quad (3.1)$$

One may use [11, Lemma 3.2] over $\mathbb{Q}$. The representation is self-contragredient. That statement first includes prime powers; Deligne’s bound removes their good contribution with $O(\sqrt{X} \log^2(2X))$ error, and the finitely many bad primes are harmless at fixed f . Decreasing c_f absorbs both errors. No conjectural prime-splitting assertion is needed.

Partial summation with $(\log X)/X$ gives

$$V_f(H) := \sum_{\substack{h_p < H \\ p \nmid N}} b_p^2 = \frac{H^2}{2} + O_f(1), \quad H \geq 0. \quad (3.2)$$

The bounded error, rather than just $o(H^2)$, follows because $(1+H)e^{-c_f\sqrt{H}}$ is integrable on the positive half-line. Endpoint atoms are covered by $b_p^2 \leq 4(\log p)^2/p$. The same calculation on a logarithmic interval $[H, H+\Delta]$, with $0 \leq H < H+\Delta \leq \ell$, gives the conservative bound

$$\sum_{H \leq h_p < H+\Delta} b_p^2 = \int_H^{H+\Delta} v \, dv + O_f((\ell+1)^2 e^{-c_f\sqrt{H}}), \quad (3.3)$$

after decreasing c_f if necessary and treating the finite initial interval by its constant. This form is valid even when $\Delta > 1$. It follows by applying Stieltjes partial summation at the two endpoints and integrating the error over the interval; the bound $(\ell+1)(1+\Delta) \leq (\ell+1)^2$ suffices.

Two further consequences used repeatedly are

$$\ell^{-2} \sum_{h_p < \ell} b_p^2 \delta_{h_p/\ell} \implies a \, da \quad \text{on } [0, 1], \quad \sum_p b_p^4 < \infty. \quad (3.4)$$

The fourth-power sum has an absolute majorant independent of f ; also $|b_p| \leq 2(\log p)/\sqrt{p} \leq 4/e$.

A relative tracial Taylor estimate. We record the noncommutative estimate that will justify changing local variables without paying the exponentially large size of their background. In a unital C^* -algebra with positive normalized trace τ , let H, Z be bounded selfadjoint elements. Duhamel's ordered-simplex formula and tracial Hölder give

$$\left| \frac{d^r}{dt^r} \tau(e^{H+tZ}) \right| \leq \|Z\|^r \tau(e^{H+tZ}). \quad (3.5)$$

The formula follows in the finite tracial von Neumann completion: each product of exponential factors has Hölder exponents whose reciprocals sum to one; the derivative factorial cancels the simplex volume. The same argument for the first derivative gives

$$\tau(e^{H+tZ}) \leq e^{t\|Z\|} \tau(e^H).$$

Here monotonicity is a statement about the traced exponential, not operator monotonicity of the exponential map. Taylor's remainder after degree $r-1$ is consequently bounded by $\|Z\|^r e^{\|Z\|} \tau(e^H)/r!$.

Let z be uniform on the circle, let S_m be contractions, and let (b_m) be a finite real string indexed by positive integers. Put

$$X(z) = \sum_m b_m (z^m S_m + (\bar{z})^m S_m^*), \quad c = \sum_m |b_m| \leq 4.$$

Let Y retain the first harmonic, or be zero. If $d = \sum_{m \text{ omitted}} b_m^2$, the linear expectations vanish, and unequal harmonics are orthogonal in the quadratic term. For the remaining terms set

$$B_H(S) = \int_0^1 \tau(e^{(1-v)H} S e^{vH} S^*) \, dv.$$

Positivity and Hölder give $0 \leq B_H(S) \leq \tau(e^H)$. The two opposite orientations cancel the Taylor factor $1/2$; their integrals agree by cyclicity. Thus the difference of the quadratic expectations is exactly $s^2 \sum_{m \text{ omitted}} b_m^2 B_H(S_m)$. Charging the two cubic remainders yields, for real s ,

$$\left| \mathbb{E} \tau(e^{H+sX}) - \mathbb{E} \tau(e^{H+sY}) \right| \leq \left(s^2 d + \frac{8}{3} |s|^3 e^{8|s|} c^3 \right) \tau(e^H). \quad (3.6)$$

Both expectations are at least $\tau(e^H)$ by convexity and centering. Therefore this is a relative estimate in both directions, uniform in the background H .

Every prime power, with its correct phase. Give each prime an independent circle variable z_p , and replace its local string by

$$X_p(z_p) = \sum_{m \log p < \ell} b_f(p^m) (z_p^m S_{m \log p} + (\overline{z_p})^m S_{-m \log p}).$$

Unique factorization says that zero total displacement is equivalent to zero signed exponent at each prime. The trace of every other word is zero. The norm-convergent exponential series at fixed ℓ therefore gives the exact identity

$$\mathbb{E} \tau_\ell \left(e^{s \sum_p X_p(z_p)} \right) = \tau_\ell(e^{s W_{f, \ell}}). \quad (3.7)$$

All harmonics at p use the same z_p . Higher-order resonances among these harmonics are retained, not assumed to vanish.

For good p retain its first harmonic; for bad p retain zero. The local budgets satisfy

$$c_p \leq \frac{2 \log p}{\sqrt{p} - 1} \leq 4, \quad \sum_p c_p^3 \leq C_* := 8 \sum_p \frac{(\log p)^3}{(\sqrt{p} - 1)^3} < \infty, \quad (3.8)$$

$$\sum_p d_p \leq D_* := 4 \sum_p \frac{(\log p)^2}{p(p-1)} < \infty. \quad (3.9)$$

At a bad prime its full squared mass is at most $(\log p)^2/(p^2 - 1)$, so (3.9) holds uniformly even as the ramified set varies. Conditioning on all other primes in (3.6) and replacing one local string at a time proves

$$\left| P_{f, \ell}(s) - \log \tau_\ell(e^{s W^{(1)}}) \right| \leq s^2 D_* + \frac{8}{3} |s|^3 e^{8|s|} C_*. \quad (3.10)$$

This is bounded independently of f and ℓ at bounded s , and tends to zero at $s = t/\ell$ for fixed t . It is not a bound for $\|W - W^{(1)}\|$.

4. THE WEIGHTED AUTOCORRELATION IDENTITY

For $u \in L^2(0, 1)$ with $\|u\|_2 = 1$ put

$$F(x) = |u(x)|^2, \quad c_u(a) = \int_0^{1-a} \overline{u(x)} u(x+a) dx, \quad J(u) = \int_0^1 a |c_u(a)|^2 da.$$

Let $\hat{F}(n) = \int_0^1 F(x) e^{-2\pi i n x} dx$. We will use the exact identity

$$\begin{aligned} \frac{1}{12} - J(u) &= \frac{1}{2\pi^2} \sum_{n \geq 1} \frac{|\hat{F}(n)|^2}{n^2} \\ &\quad + \frac{1}{2} \int_0^1 a \int_0^{1-a} \int_0^{1-a} \left| \overline{u(x)} u(x+a) - \overline{u(y)} u(y+a) \right|^2 dx dy da. \end{aligned} \quad (4.1)$$

Both terms on the right are nonnegative.

To derive it, write $g_a(x) = \overline{u(x)} u(x+a)$. The elementary variance identity on an interval of length $1-a$ is

$$(1-a) \int |g_a|^2 - \left| \int g_a \right|^2 = \frac{1}{2} \iint |g_a(x) - g_a(y)|^2 dx dy.$$

The required integrations are legitimate without assuming $u \in L^4$. Indeed Tonelli gives

$$\int_0^1 a(1-a) \int_0^{1-a} F(x) F(x+a) dx da = \frac{1}{2} \int_0^1 \int_0^1 |x-y|(1-|x-y|) F(x) F(y) dx dy < \infty.$$

The periodic Bernoulli identity

$$v(1-v) = \frac{1}{6} - \frac{1}{\pi^2} \sum_{n \geq 1} \frac{\cos(2\pi n v)}{n^2}, \quad 0 \leq v \leq 1,$$

converges absolutely and uniformly. Integrating it against $F(x)F(y)$ evaluates the preceding expression as $1/12 - (2\pi^2)^{-1} \sum_{n \geq 1} |\hat{F}(n)|^2/n^2$. Subtracting $J(u)$ proves (4.1).

Equality and what the deficit actually controls. Equality first forces $\widehat{F}(n) = 0$ for $n \neq 0$, hence $F = 1$ almost everywhere by Fourier uniqueness in L^1 . The second term then gives, for almost every a ,

$$\overline{u(x)}u(x+a) = \chi(a) \quad \text{for almost every } x \in (0, 1-a), \quad |\chi(a)| = 1.$$

The zero-extended L^2 autocorrelation is continuous, so $\chi(a) = c_u(a)/(1-a)$ is continuous on $[0, 1)$. Continuity of translations in L^2 extends the increment identity to every admissible a on compact interior intervals. Composing two increments gives $\chi(a+b) = \chi(a)\chi(b)$ for $a, b \geq 0$, $a+b < 1$. A continuous local circle character is $e^{i\omega a}$: choose a continuous argument near zero, apply the additive law there, and subdivide an arbitrary a into small equal increments. It follows that $e^{-i\omega x}u(x)$ has zero distributional derivative in $(0, 1)$ and is constant almost everywhere. Thus

$$J(u) = \frac{1}{12} \iff u(x) = e^{i\theta + i\omega x} \text{ a.e.}, \quad \theta, \omega \in \mathbb{R}. \quad (4.2)$$

If $J(u) \geq 1/12 - \varepsilon$, the first deficit is at most ε , giving $\sum_{n \geq 1} |\widehat{F}(n)|^2/n^2 \leq 2\pi^2\varepsilon$. This directly controls a negative-Sobolev measure of the squared modulus. A stronger conclusion requires the following additional compactness and local-coercivity arguments.

Strong stability modulo affine phase. For arbitrary $u \in L^2(0, 1)$ write $D(u) = \|u\|_2^4/12 - J(u) \geq 0$, and for unit u let

$$\mathcal{E} = \{e^{i\theta + i\omega x} : \theta, \omega \in \mathbb{R}\}, \quad d(u) = \inf_{e \in \mathcal{E}} \|u - e\|_2.$$

The functional J is a continuous real quartic polynomial on the complex Hilbert space viewed as a real space. This follows from $|c_u(a)| \leq \|u\|_2^2$ and polarization. Modulation and constant phase preserve D and d .

Compactness modulo symmetries followed by transverse second variation is a standard strategy in stability theory for sharp Fourier inequalities; see, for example, Christ [8]. Here a compact correlation splitting and explicit primitive identities give a direct proof for this weighted functional.

We first show that $D(u_n) \rightarrow 0$ for unit u_n forces $d(u_n) \rightarrow 0$. The essential weak-splitting fact is

$$u_n \rightharpoonup u, \quad r_n = u_n - u \implies J(u_n) = J(u) + J(r_n) + o(1). \quad (4.3)$$

In the correlation expansion, each mixed term is a compact linear or antilinear map of r_n from L_x^2 to L_a^2 . For example its kernel is $u(y-a)\mathbf{1}_{\{0 < a < y < 1\}}$, of squared Hilbert–Schmidt norm $\int_0^1 (1-x)|u(x)|^2 dx \leq \|u\|_2^2$. These terms tend strongly to zero. The remaining cross term is

$$\int_{0 < x < y < 1} (y-x)c_u(y-x)r_n(x)\overline{r_n(y)} dx dy.$$

Its kernel is bounded, hence Hilbert–Schmidt, so it also tends to zero. This proves (4.3); it does not assume the self-correlation of a weak-null sequence tends to zero.

Plancherel for the zero-extended profile gives

$$J(u_n) \leq \frac{1}{2} \int_{\mathbb{R}} |c_{u_n}(a)|^2 da = \frac{1}{4\pi} \int_{\mathbb{R}} |\widehat{u_n}(\xi)|^4 d\xi \leq \frac{1}{2} \|\widehat{u_n}\|_{\infty}^2.$$

Thus a near-maximizing sequence can be modulated and rotated so that its mean is real and bounded below by a positive constant. A weakly convergent subsequence then has a nonzero limit u . If $\alpha = \|u\|_2^2$, weak orthogonality gives $\|r_n\|_2^2 \rightarrow 1 - \alpha$. By (4.3) and the homogeneous sharp inequality,

$$\frac{1}{12} \leq \frac{\alpha^2 + (1-\alpha)^2}{12}.$$

Since $0 < \alpha \leq 1$, necessarily $\alpha = 1$. The convergence is strong, and its limit belongs to $\mathcal{E}$ by (4.2). This proves the claimed compactness modulo modulation.

The local estimate has an explicit sharp quadratic constant. Write $h = a + ib$ with a, b real, $\mu = \int_0^1 a$, and put

$$L_h(r) = \int_0^{1-r} (\overline{h(x)} + h(x+r)) dx.$$

Exact expansion at the flat profile gives

$$D(1+h) = Q(h) + C_3(h) + D(h), \quad |C_3(h)| \leq \frac{7}{3} \|h\|_2^3, \quad (4.4)$$

where

$$\begin{aligned} Q(h) &= \frac{\|h\|_2^2}{6} + \frac{\mu^2}{3} - \int_0^1 r |L_h(r)|^2 dr - 2 \int_0^1 r(1-r) \Re c_h(r) dr, \\ C_3(h) &= \frac{\mu \|h\|_2^2}{3} - 2 \Re \int_0^1 r L_h(r) \overline{c_h(r)} dr. \end{aligned} \quad (4.5)$$

The remainder bound follows from $|L_h(r)| \leq 2\|h\|_2$ and $|c_h(r)| \leq \|h\|_2^2$. Since $D \geq 0$, Q is nonnegative. Its zero directions contain $\mathcal{K} = \text{span}_{\mathbb{R}}\{1, i, ix\}$ by scaling, phase, and modulation invariance. Positivity makes all bilinear cross terms with $\mathcal{K}$ vanish.

In fact the transverse quadratic gap is exactly $1/6$. If $h = a + ib \perp \mathcal{K}$ in the real Hilbert inner product, then $\int a = \int b = \int xb = 0$. Let $A(x) = \int_0^x a$ and $B(x) = \int_0^x b$. Use subscripts e, o for even and odd parts under reflection $x \mapsto 1-x$, and write $\overline{A} = \int_0^1 A$. Both primitives vanish at the endpoints, and

$$\Re L_h(r) = A(1-r) - A(r) = -2A_o(r), \quad \Im L_h(r) = -B(r) - B(1-r) = -2B_e(r).$$

For a real mean-zero function v with primitive V , the same Bernoulli expansion used above gives

$$-2 \int_0^1 r(1-r) c_v(r) dr = 2 \left(\|V\|_2^2 - \left| \int_0^1 V \right|^2 \right).$$

This also follows from Parseval, since the nonzero Fourier coefficients of V are those of v divided by $2\pi in$. The square of either reflection-parity component is symmetric, so its integral against r is half its integral. Substitution in (4.5), using $\int B = -\int xb = 0$, yields

$$Q(a) = \frac{\|a\|_2^2}{6} + 2\|A_e - \overline{A}\|_2^2, \quad Q(ib) = \frac{\|b\|_2^2}{6} + 2\|B_o\|_2^2. \quad (4.6)$$

There are no real-imaginary cross terms. Hence $Q(h) \geq \|h\|_2^2/6$ on $\mathcal{K}^\perp$, with equality for $h(x) = \cos(2\pi x)$.

For a unit u , a nearest affine phase exists because

$$d(u)^2 = 2 - 2 \max_{\omega \in \mathbb{R}} \left| \int_0^1 u(x) e^{-i\omega x} dx \right|,$$

and the Fourier transform is continuous, nonzero, and vanishes at infinity. Align that nearest phase with 1 and set $h = u - 1$, $\delta = \|h\|_2 = d(u)$. Differentiating the maximizing phase and frequency gives $\Im \int h = \Im \int xh = 0$; the derivatives exist since $xu \in L^1$. Normalization gives $\Re \int h = -\delta^2/2$. Thus $\|P_{\mathcal{K}} h\|_2^2 = \delta^4/4$, and (4.4)–(4.6) imply

$$D(u) \geq \delta^2 \left(\frac{1}{6} - \frac{7}{3} \delta - \frac{\delta^2}{24} \right). \quad (4.7)$$

For $\delta \leq 1/32$ this is at least $\delta^2/12$. By the compactness already proved, the infimum of $D(u)$ over unit profiles with $d(u) \geq 1/32$ is some $\eta_0 > 0$. Since $d(u)^2 \leq 2$, the remaining profiles satisfy $d(u)^2 \leq 2D(u)/\eta_0$. This proves the global bound in (1.11), with an absolute but nonexplicit constant $\max(12, 2/\eta_0)$.

As $D(u) \downarrow 0$, compactness places u in the local neighborhood, where $\delta^2 \leq 12D(u)$. Equation (4.7) then gives uniformly

$$d(u)^2 \leq 6D(u) + O(D(u)^{3/2}). \quad (4.8)$$

This coefficient cannot be reduced. For small real t take $u_t = (1 + t \cos(2\pi x)) / \sqrt{1 + t^2/2}$. Positivity and the triangle inequality make frequency zero maximize its Fourier modulus, so its nearest phase is 1. Then $d(u_t)^2 \sim t^2/2$, whereas (4.4) and (4.6) give $D(u_t) \sim t^2/12$. Their ratio tends to 6. In particular the square-root distance exponent is optimal. We make no claim that 6 is the sharp global constant.

The finite-dimensional identity. For a unit vector $v = (v_0, \dots, v_{d-1})$ define $F_i = |v_i|^2$, $c_j = \sum_{i=0}^{d-j-1} \overline{v_i} v_{i+j}$, and $\widehat{F}_d(k) = \sum_{i=0}^{d-1} F_i e^{-2\pi i k i/d}$. The cyclic kernel $K_d(r) = r(d-r)$ has normalized Fourier coefficients

$$\widehat{K}_d(0) = \frac{d^2 - 1}{6}, \quad \widehat{K}_d(k) = -\frac{1}{2 \sin^2(\pi k/d)}, \quad 1 \leq k < d.$$

For example these follow by applying the cyclic second difference: $K_d(r+1) - 2K_d(r) + K_d(r-1) = -2 + 2d\mathbf{1}_{\{r=0\}}$. The same variance calculation on each overlap yields

$$\begin{aligned} \frac{d^2 - 1}{12} - \sum_{j=1}^{d-1} j |c_j|^2 &= \frac{1}{4} \sum_{k=1}^{d-1} \frac{|\widehat{F}_d(k)|^2}{\sin^2(\pi k/d)} \\ &\quad + \frac{1}{2} \sum_{j=1}^{d-1} j \sum_{i,h=0}^{d-j-1} |\overline{v_i} v_{i+j} - \overline{v_h} v_{h+j}|^2. \end{aligned} \quad (4.9)$$

Only the kernel on the moduli has been evaluated cyclically; c_j is still the aperiodic, killed autocorrelation. In particular

$$\sum_{j=1}^{d-1} j |c_j|^2 \leq \frac{d^2 - 1}{12}. \quad (4.10)$$

For $d \geq 2$, equality forces $F_i = 1/d$, and equality at lag one forces all successive phase increments to agree. Its complete equality class is $v_i = d^{-1/2} e^{i\theta + i\omega i}$. The $d = 1$ case is immediate.

Equivalently, every nonnegative trigonometric polynomial Q of degree at most n and mean one satisfies

$$\sum_{k \in \mathbb{Z}} |k| |\widehat{Q}(k)|^2 \leq \frac{n(n+2)}{6}.$$

Fejér–Riesz factorization gives this formulation from (4.10); its extremizers for $n \geq 1$ are translated Fejér kernels. This is a useful equivalent description of the same inequality, not an additional independent extremal result.

For comparison with a weighted linear positive-definite extremum, $\Lambda(a) = c_u(a)$ is positive definite and supported in $[-1, 1]$, and $|\Lambda|^2$ is positive definite as well. Enlarging the class of $|\Lambda|^2$ to all normalized positive-definite functions loses the square constraint. That larger class contains $\phi(a) = (1 - |a|)_+$, for which $\int_{-1}^1 |a| \phi(a) da = 1/3$; our quadratic optimum is $\int_{-1}^1 |a| (1 - |a|)^2 da = 1/6$. Thus the direct application of a linear weighted theorem such as [15] cannot give the constant in (1.10).

5. UNIFORM PRESSURE THROUGH A BOUNDARY-PRESERVING GRID

We prove (1.3). Fix $C > 0$ and take $2 \leq t \leq C\ell$; negative t will follow at the end. All sums in this section initially contain only good first primes.

The flat lower bound. After prime-phase expectation the exponential operator is a multiplication operator: exactly the primewise balanced words survive, and every such word is multiplication by its full survival indicator. Its coefficients are nonnegative products of even powers of the real b_p . Taking the flat unit vector $v = \ell^{-1/2} \mathbf{1}_{I_\ell}$ and applying scalar Jensen to its spectral measure gives

$$\tau_\ell(e^{tW^{(1)}/\ell}) \geq \prod_{h_p < \ell} I_0 \left(\frac{2tb_p}{\ell} (1 - h_p/\ell) \right), \quad (5.1)$$

where $I_0(2x) = \sum_{r \geq 0} x^{2r}/(r!)^2$. For bounded $|x|$, $\log I_0(2x) = x^2 + O(x^4)$. Using (3.2) and integration by parts,

$$\sum_{h_p < \ell} b_p^2 (1 - h_p/\ell)^2 = \frac{2}{\ell} \int_0^\ell V_f(H) (1 - H/\ell) dH = \frac{\ell^2}{12} + O_f(1).$$

The fourth-power remainder is $O_C(t^4/\ell^4) = O_C(1)$. This proves the lower side of (1.3) for $W^{(1)}$, with a bounded error throughout the stated range.

Word truncation and rounding. Choose once and for all $D > 0$ with $D \log 2 > 4$, and set

$$M = \lceil Dt^2 \rceil, \quad q = \lceil t^6 \rceil, \quad d = q + 2M, \quad j_p = \lfloor qh_p/\ell \rfloor. \quad (5.2)$$

Here $0 \leq j_p < q$; the zero bin must be retained. On $\mathbb{C}^d$ use the killed integer shifts, with $S_0 = I_d$, and give each prime its own independent circle phase, even when several primes have the same j_p . Put

$$B_z = \sum_{h_p < \ell} \frac{b_p}{\ell} (z_p S_{j_p} + \bar{z}_p S_{j_p}^*).$$

A balanced word of length $r \leq M$ has original rescaled range R and integer-grid range R_q . Each partial sum moves by at most r/q , hence $|R_q/q - R| \leq 2r/q$. It is exactly closed on both grids, and

$$(1 - R)_+ \leq \frac{d}{q} (1 - R_q/d)_+. \quad (5.3)$$

Indeed $d - R_q \geq q(1 - R) + 2(M - r)$. Unbalanced arithmetic words which close after rounding still vanish under the independent prime phases.

The positive first-prime series has tail above degree M at most

$$2^{-M} \prod_p I_0(4tb_p/\ell) \leq \exp\{-M \log 2 + 4t^2 V_f(\ell)/\ell^2\} \leq e^{-t^2}$$

for all sufficiently large ℓ , uniformly in $2 \leq t \leq C\ell$. We used $I_0(2x) \leq e^{x^2}$ and $V_f(\ell)/\ell^2 \rightarrow 1/2$. Consequently

$$\tau_\ell(e^{tW^{(1)}/\ell}) \leq \frac{d}{q} \mathbb{E}_z \operatorname{tr}_d(e^{tB_z}) + e^{-t^2}, \quad \operatorname{tr}_d = \operatorname{Tr}/d. \quad (5.4)$$

This multiplicative comparison is uniform at the exponential coupling being estimated; it is not a fixed-moment approximation.

Relative replacement by circular Gaussians. Replace z_p successively by independent $\zeta_p = (g_{p,1} + ig_{p,2})/\sqrt{2}$, where the $g_{p,a}$ are real standard Gaussians. Denote the resulting matrix by G . Circle and circular Gaussian variables have the same mixed moments through degree three. For a Hermitian background H , the fourth-order version of (3.5) bounds the Taylor remainder by $\|Z\|^4 e^{\|Z\|} \operatorname{Tr}(e^H)/24$. With $Z = (tb_p/\ell)(zS + \bar{z}S^*)$, its expected value is at most $C_C (tb_p/\ell)^4 \operatorname{Tr}(e^H)$, for either distribution. This follows from $\|Z\| \leq 2|tb_p/\ell||z|$ and the finite Gaussian exponential moments; $|tb_p/\ell|$ is bounded uniformly by a constant depending only on C . Both centered expectations are at least $\operatorname{Tr}(e^H)$. As in the local restoration argument, comparison is relative and gives

$$\left| \log \mathbb{E}_z \operatorname{tr}_d(e^{tB_z}) - \log \mathbb{E} \operatorname{tr}_d(e^{tG}) \right| \leq C_C \frac{t^4}{\ell^4} \sum_p b_p^4 = O_C(1). \quad (5.5)$$

There is no dimension-dependent denominator or norm bound on the background in this estimate.

The two Gaussian variances. Write $G = \sum_a g_a A_a$. Direct multiplication gives

$$\sum_a A_a^2 = \ell^{-2} \sum_p b_p^2 (S_{j_p} S_{j_p}^* + S_{j_p}^* S_{j_p}) \leq 2V_f(\ell) \ell^{-2} I_d.$$

The Gaussian matrix-series bound [19] therefore yields

$$\mathbb{E} \lambda_{\max}(G) \leq 2\sqrt{V_f(\ell) \ell^{-2} \log d} = O_f(\sqrt{\log(2+t)}). \quad (5.6)$$

For completeness, its trace-mgf estimate follows by Gaussian integration by parts and

$$\mathrm{Tr}(Ae^{vB} A e^{(1-v)B}) \leq \mathrm{Tr}(A^2 e^B), \quad 0 \leq v \leq 1,$$

for Hermitian A, B , which is scalar convexity in an eigenbasis of B . Thus $\mathbb{E} \mathrm{Tr} e^{\theta G} \leq d \exp(\theta^2 \|\sum A_a^2\|/2)$; optimizing in θ proves (5.6).

The sharp quadratic coefficient uses a different quantity:

$$\sigma_w^2 = \sup_{\|v\|=1} \sum_a \langle v, A_a v \rangle^2 = \frac{2}{\ell^2} \sup_{\|v\|=1} \sum_p b_p^2 |c_{j_p}(v)|^2, \quad c_0(v) = 1. \quad (5.7)$$

The top eigenvalue is the supremum of its real linear Rayleigh functionals, and so is σ_w -Lipschitz in the real Gaussian variables. Gaussian logarithmic Sobolev concentration [12, 14] gives

$$\log \mathbb{E} e^{t \lambda_{\max}(G)} \leq t \mathbb{E} \lambda_{\max}(G) + t^2 \sigma_w^2 / 2. \quad (5.8)$$

In detail, for a Lipschitz function F with constant σ_w , the logarithmic Sobolev inequality applied to $e^{tF/2}$ gives $t\Psi'(t) - \Psi(t) \leq t^2 \sigma_w^2 / 2$, where $\Psi(t) = \log \mathbb{E} e^{tF}$; integrate from zero. Smooth approximation proves the assertion for the top eigenvalue.

All grid vectors and all intermediate couplings. Let $H_0 = B_f \log^2(2 + \ell)$, where B_f is large enough that $c_f \sqrt{B_f} > 12$, and take ℓ so large that $H_0 < \ell$. Write $\Delta = \ell/q$. All bins whose left endpoint is less than H_0 have total normalized mass at most

$$\ell^{-2} V_f(H_0 + \Delta) = O_f \left(\frac{\log^4(2 + \ell)}{\ell^2} + q^{-2} \right). \quad (5.9)$$

This includes the scalar zero bin and any straddling bin. For the remaining bins, (3.3) and $q = O_C(\ell^6)$ show that the sum of all normalized errors is $O_{f,C}(\ell^{-6})$. Their main masses are $(j + 1/2)/q^2$. Since every $|c_j(v)| \leq 1$, the error bounds hold for all unit vectors simultaneously. Enlarge the positive main sum to $1 \leq j < d$ and apply (4.10). We obtain

$$\begin{aligned} \sigma_w^2 &\leq \frac{2}{q^2} \left(\frac{d^2 - 1}{12} + \frac{d}{2} \right) + O_{f,C} \left(\frac{\log^4(2 + \ell)}{\ell^2} + q^{-2} + \ell^{-6} \right) \\ &= \frac{1}{6} + O_{f,C} \left(\frac{M}{q} + \frac{1}{q} + \left(\frac{M}{q} \right)^2 + \frac{\log^4(2 + \ell)}{\ell^2} \right). \end{aligned} \quad (5.10)$$

The extra q^{-2} and ℓ^{-6} terms are absorbed in the last line. This is where weak convergence of the variance measures would have been insufficient: the maximizing vector changes with the grid, and all its bins must be controlled together.

Since $\mathrm{tr}_d(e^{tG}) \leq e^{t \lambda_{\max}(G)}$, equations (5.6)–(5.10) give

$$\log \mathbb{E} \mathrm{tr}_d(e^{tG}) \leq \frac{t^2}{12} + O_{f,C} \left(t \sqrt{\log(2+t)} + 1 + \frac{t^2 \log^4(2 + \ell)}{\ell^2} \right).$$

Here $t^2(M/q + q^{-1} + (M/q)^2) = O_D(1)$. The last displayed error is uniformly absorbed by $O_C(t \sqrt{\log(2+t)} + 1)$ when $2 \leq t \leq C\ell$: indeed $t \log^4(2 + \ell)/\ell^2 \leq C \log^4(2 + \ell)/\ell$, which is bounded. The factor d/q , the additive tail in (5.4), and (5.5) cost only bounded amounts in the logarithm. The Gaussian trace expectation is at least one, so the additive tail creates no denominator difficulty.

Finally (3.10) restores the complete local strings at coupling $t/\ell \leq C$, again at bounded cost. The first-prime trace is even in t , since replacing every z_p by $-z_p$ changes its sign. Apply

the same restoration for negative t . Together with (5.1), this proves (1.3) in its full range, and hence (1.4) and (1.6).

6. INTERMEDIATE SCALES AND BOUNDARY SURVIVAL

A scalar large-deviation argument. Let $r_\ell \rightarrow \infty$ with $r_\ell = O(\ell)$, and let μ_ℓ be the probability spectral law of $W_{f,\ell}/(\ell r_\ell)$ in τ_ℓ . Equation (1.3) gives, for every fixed $s \in \mathbb{R}$,

$$\frac{1}{r_\ell^2} \log \int e^{sr_\ell^2 x} d\mu_\ell(x) = \frac{1}{r_\ell^2} P_{f,\ell}(sr_\ell/\ell) \rightarrow \frac{s^2}{12}. \quad (6.1)$$

For $s \neq 0$ one applies the uniform estimate at $t = |s|r_\ell$ with its fixed upper constant enlarged by $|s|$; $s = 0$ is immediate. This works however slowly r_ℓ diverges.

We give the short scalar tilting proof to specify the measure and the quantifiers. The same argument will later apply to the central limiting law. More generally let probability measures μ_j have speed $v_j \rightarrow \infty$ and scaled logarithmic mgf $L_j(s) = \log \int e^{sv_j x} d\mu_j(x)$ with $L_j(s)/v_j \rightarrow s^2/12$ for every real s . Define

$$d\mu_{j,s}(x) = e^{sv_j x - L_j(s)} d\mu_j(x).$$

For $h > 0$, exponential Markov gives

$$\limsup_j v_j^{-1} \log \mu_{j,s}(x \geq s/6 + \eta) \leq \frac{(s+h)^2 - s^2}{12} - h(s/6 + \eta) = h^2/12 - h\eta.$$

Take $h = 6\eta$. A negative increment proves the corresponding left-tail estimate. Thus

$$\limsup_j v_j^{-1} \log \mu_{j,s}(|x - s/6| \geq \eta) \leq -3\eta^2. \quad (6.2)$$

For any $a \in \mathbb{R}$, set $s = 6a$ and undo the tilt on $(a - \eta, a + \eta)$; its tilted mass tends to one. This gives

$$\liminf_j v_j^{-1} \log \mu_j((a - \eta, a + \eta)) \geq -3a^2 - 6|a|\eta.$$

Shrinking the interval inside an open set proves the lower bound. For an upper bound near a use Markov with $s = 6a$; a finite cover gives the bound on compact sets. Markov at $s = \pm 6R$ also gives exponential tightness, $\limsup_j v_j^{-1} \log \mu_j(|x| \geq R) \leq -3R^2$. It extends the compact upper bound to every closed set. Consequently for every Borel set B ,

$$-\inf_{B^\circ} 3x^2 \leq \liminf_j v_j^{-1} \log \mu_j(B) \leq \limsup_j v_j^{-1} \log \mu_j(B) \leq -\inf_{\overline{B}} 3x^2. \quad (6.3)$$

We use $\log 0 = -\infty$ and $\inf \emptyset = +\infty$. Applying this proof to (6.1) establishes the claimed intermediate-scale principle. For every fixed $a > 0$,

$$\begin{aligned} \lim_{\ell \rightarrow \infty} r_\ell^{-2} \log \tau_\ell(\mathbf{1}_{[a\ell r_\ell, \infty)}(W_{f,\ell})) &= -3a^2, \\ \lim_{\ell \rightarrow \infty} r_\ell^{-2} \log \tau_\ell(\mathbf{1}_{(-\infty, -a\ell r_\ell]}(W_{f,\ell})) &= -3a^2. \end{aligned} \quad (6.4)$$

The equality follows by sandwiching between open and closed rays; their rate infima agree. Equal leading tail rates do not assert exact symmetry of the full finite-window prime-power law.

Which spectral mass carries the Weyl coefficient. Taking $r_\ell = \ell$, the exponential tilt at fixed coupling s concentrates $W_{f,\ell}/\ell^2$ at $s/6$ by (6.2). At the Weyl coupling $s = 1/2$ this value is $1/12$. One can also recover the tilted first moment without differentiating an asymptotic expansion. The functions $\ell^{-2}P_{f,\ell}(s)$ are convex and differentiable. For $h > 0$ their derivatives lie between their left and right difference quotients. First let ℓ tend to infinity in these bounds and then let $h \downarrow 0$. This proves

$$\frac{\tau_\ell(W_{f,\ell} e^{sW_{f,\ell}})}{\ell^2 \tau_\ell(e^{sW_{f,\ell}})} \rightarrow \frac{s}{6}. \quad (6.5)$$

No convergence of second derivatives, tilted variances, or tilted Gaussian fluctuations is inferred by this argument.

The exact exponential cost of surviving the boundary. For first primes and fixed $s \neq 0$, introduce independent integer counts R_p with

$$\mathbb{P}(R_p = r) = \frac{(|s||b_p|)^{2r}}{(r!)^2 I_0(2|s||b_p|)}, \quad r \geq 0.$$

Given these counts, choose uniformly an ordering of R_p copies of $+h_p$ and R_p copies of $-h_p$ for each prime. Let $F_\ell = (1 - \text{range}/\ell)_+$ be the surviving fraction of starting positions. Counting these orderings in the exponential series gives the exact formula

$$\tau_\ell(e^{sW^{(1)}}) = \prod_p I_0(2|s||b_p|) \mathbb{E}F_\ell. \quad (6.6)$$

These are paired words, not paths with independent increments. By (3.2) and the summable fourth-power error, the logarithm of the unmasked product is $s^2\ell^2/2 + O_{f,s}(1)$. The pressure estimate gives

$$\log \mathbb{E}F_\ell = -\frac{5s^2}{12}\ell^2 + O_{f,s}(\ell\sqrt{\log(2+\ell)}). \quad (6.7)$$

Thus the boundary contributes at leading exponential order, not merely as a lower-order endpoint correction.

7. THE CENTRAL ARITHMETIC LAW AND ITS SHARP TAILS

A complete moment description. For a pair partition π of $\{1, \dots, 2m\}$ label its pairs in order of first occurrence. Give pair j a length $a_j \in [0, 1]$ and an orientation $\varepsilon_j \in \{1, -1\}$. Its first occurrence contributes increment $\varepsilon_j a_j$ and its second contributes $-\varepsilon_j a_j$. If $r_0 = 0, r_1, \dots, r_{2m} = 0$ are the partial sums, put

$$F_{\pi, \varepsilon}(a) = (1 - \max_i r_i + \min_i r_i)_+.$$

The central moments are

$$m_0 = 1, \quad m_{2m+1} = 0, \quad m_{2m} = \sum_{\pi} \sum_{\varepsilon \in \{\pm 1\}^m} \int_{[0,1]^m} F_{\pi, \varepsilon}(a) \prod_{j=1}^m a_j da_j. \quad (7.1)$$

In particular $0 \leq m_{2m} \leq (2m-1)!!$, since the masks are at most one and $2 \int_0^1 a da = 1$. The formal exponential series therefore satisfies

$$M(s) := \sum_{m \geq 0} \frac{m_{2m} s^{2m}}{(2m)!} \leq e^{s^2/2} \quad (s \in \mathbb{R}). \quad (7.2)$$

We obtain existence of its probability measure from the actual operators. Expand $\tau_\ell((W^{(1)}/\ell)^r)$. Every surviving word is balanced at each prime, so odd moments vanish. At degree $2m$, words with exactly m distinct primes, each occurring twice, are indexed by π, ε , and distinct prime labels. Their masks are exactly those in (7.1). Write $c_{p,\ell} = b_p/\ell$. Allowing repeated labels in this pairing formula, and bounding words with a prime used four or more times, costs at most

$$C_m \left(\sum_p c_{p,\ell}^4 \right) \left(1 + \sum_p c_{p,\ell}^2 \right)^{m-2} = o_f(1) \quad (m \geq 2).$$

To see this, select two paired labels with a common prime, sum that prime with fourth-power weight, and sum the other labels freely. There are only finitely many overcountings at fixed m , and every mask is bounded by one. For $m = 1$ there is no collision error.

The masks are continuous in their lengths. The variance-measure convergence in (3.4) now gives exactly (7.1). Also the unmasked positive-word bound gives, for every real s ,

$$\tau_\ell(e^{sW^{(1)}/\ell}) \leq \prod_p I_0(2sb_p/\ell) \leq \exp(s^2 V_f(\ell)/\ell^2). \quad (7.3)$$

Second moments give tightness of the spectral laws. Higher moments give uniform integrability of each fixed power, and (7.3) at larger positive and negative arguments gives uniform integrability of each fixed exponential. Every subsequential limit has (7.1), and the Gaussian moment majorant makes this moment problem determinate. Hence there is a unique probability measure ν , with mgf M , and the whole sequence converges weakly, in moments, and in real exponential moments.

Applying (3.10) at s/ℓ shows that the full and first-prime logarithmic mgfs differ by $O_s(\ell^{-2})$. It also provides uniform exponential bounds at all larger fixed arguments for the full operators. Tightness, subsequence uniqueness, and uniform integrability then prove the same three convergences for $W_{f,\ell}/\ell$. All local factors are included.

The Toeplitz identification and the low moments. For the matrices (1.7), the circular Gaussian Wick formula pairs opposite orientations. A closed integer word has normalized trace $(1 - \text{range}/n)_+$. The variance of its j th diagonal is j/n^2 , and the sums over j/n are Riemann sums. The expected normalized trace moments therefore converge to (7.1). Uniform unmasked Gaussian bounds justify all fixed exponential moments as well. This identifies ν as the expected limiting spectral measure; no almost-sure assertion is needed.

This law is the $\sqrt{a}$ specialization of the variance-profile theorem in [4], Theorem 3.3 and Section 7.2.2 in the arXiv version cited there. Their real symmetric formulation has the same limiting moments. Indeed a real Wick pair with equal rather than opposite orientations imposes a nontrivial linear closure equation on the diagonal lengths. At fixed order this leaves $O(n^{m-1})$ choices rather than $O(n^m)$; each variance is at most $1/n$, so its normalized contribution is $O(n^{-1})$. Coincident diagonal lengths have the same lower-order counting loss. The opposite orientations give exactly the circular complex limit above.

At order two the two orientations each have mask $1 - a$, giving $m_2 = 2 \int_0^1 a(1 - a) da = 1/3$. At order four set

$$I_{\max} = \int_0^1 \int_0^1 ab(1 - \max(a, b)) da db = \frac{1}{20}, \quad I_{\Sigma} = \int_{a+b \leq 1} ab(1 - a - b) da db = \frac{1}{120}.$$

Splitting the square into its two triangles proves the first evaluation; integrating in b first proves the second. For each of the pair words $AABB$ and $ABBA$, two orientations give $I_{\max}$ and two give I_{Σ} . The word $ABAB$ has four orientations giving I_{Σ} . Therefore

$$m_2 = \frac{1}{3}, \quad m_4 = 4I_{\max} + 8I_{\Sigma} = \frac{4}{15}. \quad (7.4)$$

These numbers are evaluations of the specified profile law, not separate general random-matrix assertions.

A direct large-coupling estimate for the limiting law. Taking $\ell \rightarrow \infty$ at fixed s in the flat-vector lower bound (5.1) gives

$$M(s) \geq e^{s^2/12} \quad (s \in \mathbb{R}). \quad (7.5)$$

For a matching upper estimate work directly with (7.1), not with a diagonal substitution in a fixed- s convergence statement. By symmetry take $s \geq 2$. Choose $L = \lceil Ds^2 \rceil$, $q = \lceil s^6 \rceil$, $d = q + 2L$, with $D \log 2 > 4$. The part of the positive moment series above degree L is at most $2^{-L}M(2s) \leq e^{-s^2}$ by (7.2). Round each length a down to j/q . The same enlarged-interval argument as in (5.3) dominates every retained mask by d/q times its grid mask. The bin masses now are exact:

$$w_j = \int_{j/q}^{(j+1)/q} a da = \frac{j + 1/2}{q^2}, \quad 0 \leq j < q.$$

Let

$$G_{q,d} = \sum_{j=0}^{q-1} \sqrt{w_j} (z_j S_j + \bar{z}_j S_j^*), \quad S_0 = I_d.$$

The Wick expansion of its exponential is precisely the rounded pairing expansion, including multiple pairs in one bin and the scalar zero-lag term. Thus

$$M(s) \leq \frac{d}{q} \mathbb{E} \operatorname{tr}_d(e^{sG_{q,d}}) + e^{-s^2}. \quad (7.6)$$

The strong real-Gaussian variance is at most $2 \sum_j w_j I_d = I_d$, so $\mathbb{E} \lambda_{\max}(G_{q,d}) \leq \sqrt{2 \log d}$. By (4.10) its weak variance is bounded by

$$\frac{2}{q^2} \left(\frac{d^2 - 1}{12} + \frac{d}{2} + \frac{1}{2} \right) = \frac{1}{6} + O(L/q + q^{-1} + (L/q)^2).$$

The Gaussian concentration calculation now gives

$$\frac{s^2}{12} \leq \log M(s) \leq \frac{s^2}{12} + C|s| \sqrt{\log(2 + |s|)} \quad (|s| \geq s_0), \quad (7.7)$$

with absolute constants. The grid errors multiplied by s^2 are bounded, d/q is bounded, and the additive tail in (7.6) is harmless because the centered Gaussian trace expectation is at least one.

If Y has law ν , then $R^{-2} \log \mathbb{E} e^{sR^2(Y/R)} = R^{-2} \log M(sR) \rightarrow s^2/12$. Apply the tilting proof of Section 6 with speed R^2 . Open and closed rays give exactly the two limits in (1.8). In particular both sides of the support of ν are unbounded. The quadratic logarithmic tails do not make the distribution Gaussian.

8. SEVERAL ARITHMETIC FORMS IN ONE TRANSLATION ALGEBRA

Fix pairwise inequivalent primitive holomorphic forms $f_1, \dots, f_d$, each satisfying the earlier real, trivial-character hypotheses. The tuple is fixed throughout. Let $\mathcal{P}_0$ be the primes good for every member, and put $b_i(p) = \lambda_{f_i}(p) \log p / \sqrt{p}$ there. The off-diagonal version of the same prime number theorem, [11, Theorem 1.1], gives

$$\sum_{\substack{p \leq X \\ p \in \mathcal{P}_0}} \lambda_{f_i}(p) \lambda_{f_j}(p) \log p = \delta_{ij} X + O_f(X e^{-c_f \sqrt{\log X}}). \quad (8.1)$$

The theorem excludes imaginary norm twists in its off-diagonal case. Here such an equivalence would force the twist parameter to be zero because both central characters are trivial; thus pairwise inequivalence suffices. Distinct quadratic twists qualify only when the resulting representations are genuinely inequivalent. A complex-multiplication self-twist does not create a new color.

Weighted partial summation yields $\sum_{h_p < H} b_i(p) b_j(p) = \delta_{ij} H^2/2 + O_f(1)$. The signed measures $\ell^{-2} \sum b_i(p) b_j(p) \delta_{h_p/\ell}$ converge to $\delta_{ij} a$ and have uniformly bounded total variations by Cauchy–Schwarz. Their product measures consequently converge against continuous masks: approximate a mask uniformly by sums of product functions and use the total-variation bounds.

The colored moment formula. Put $X_{i,\ell} = W_{f_i,\ell}/\ell$, all in the same algebra $\mathcal{A}_\ell$. The limit of a fixed mixed moment of even length is

$$\lim_{\ell \rightarrow \infty} \tau_\ell(X_{i_1,\ell} \cdots X_{i_{2m},\ell}) = \sum_{\substack{\pi \text{ a pairing} \\ \text{each pair has equal colors}}} \sum_{\varepsilon \in \{\pm 1\}^m} \int F_{\pi,\varepsilon}(a) \prod_{j=1}^m a_j da_j. \quad (8.2)$$

Odd limits vanish. For the common-good first-prime parts the proof is the earlier distinct-prime pairing expansion, now using the signed cross-variance measures. Repeated labels are bounded by the same fourth-power majorant, since the tuple is fixed.

To restore all local factors in mixed words, write R_i for the difference between the full operator and its common-good first-prime part. The two-letter trace gives exactly

$$\tau_\ell(R_i^2) = 2 \sum_{\text{omitted active } p^m} b_{f_i}(p^m)^2 (1 - m \log p / \ell)_+ \leq C_f.$$

Besides higher powers this includes finitely many primes bad for another tuple member. Thus $\|R_i/\ell\|_{L^2(\tau_\ell)} = O_f(\ell^{-1})$. The scalar exponential bounds give uniformly bounded tracial L^p norms at each fixed finite p for all normalized full and first-prime operators. Telescope a mixed word of length $r \geq 2$, put its remainder in L^2 , and its other $r - 1$ factors in $L^{2(r-1)}$. Tracial Hölder makes each replacement $O_{f,r}(\ell^{-1})$. The first moments are zero. This proves (8.2) for the original full operators.

Taking limits of $\tau_\ell(P^*P)$ shows that these moments define a positive normalized tracial functional on polynomials in selfadjoint variables $X_1, \dots, X_d$. It is the weighted colored Toeplitz law realized by independent circular Gaussian diagonal families on one common matrix space. This does not assert the existence of bounded limiting variables; their scalar laws have unbounded support.

For distinct colors X, Y , the pair integrals already computed give

$$\tau(XY) = 0, \quad \tau(X^2Y^2) = \frac{7}{60}, \quad \tau(XYXY) = \frac{1}{30}, \quad \tau([X, Y]^*[X, Y]) = \frac{1}{6}. \quad (8.3)$$

The last identity follows by expanding the commutator and using cyclicity. Thus orthogonality of the arithmetic prime coefficients does not produce commuting variables or freeness. Nor are the second powers classically independent: $7/60 \neq 1/9$. These profile-specific values illustrate the established colored Toeplitz phenomenon [3].

Orthogonal invariance without classical independence. For $a \in \mathbb{R}^d$ set $X(a) = \sum_i a_i X_i$. Multilinearity in (8.2) gives, for vectors $a_1, \dots, a_{2m}$,

$$\tau(X(a_1) \cdots X(a_{2m})) = \sum_{\pi} \left(\prod_{\{r,s\} \in \pi} a_r \cdot a_s \right) \sum_{\varepsilon} \int F_{\pi, \varepsilon}(v) \prod_j v_j \, dv_j. \quad (8.4)$$

Hence the entire joint moment law is invariant under real orthogonal changes of color. In particular the scalar law of $X(a)$ is the dilation of ν by $\|a\|_2$; its moments and exponential moments determine it uniquely. Orthogonal invariance here is compatible with the nonzero commutator in (8.3); it is not a multivariate classical Gaussian law.

There is also a growing-window statement for each fixed real linear combination $B_{a, \ell} = \sum_i a_i W_{f_i, \ell}$:

$$\log \tau_\ell(e^{B_{a, \ell}}) = \frac{\|a\|_2^2}{12} \ell^2 + O_{f, a}(\ell \sqrt{\log(2 + \ell)}), \quad (8.5)$$

with bounded lower error. Retain every actual first harmonic and form the real coefficient array $\sum_i a_i b_i(p)$. Equation (8.1) gives its variance density $\|a\|_2^2 a \, da$, including the moving-bin error. The positive balanced-word and Gaussian arguments then apply to this single array. The higher local harmonics are restored by (3.6), with constants depending on the fixed tuple and a . One phase is used at each prime, shared by all the forms and all its harmonics. Assigning independent phases to different forms at the same prime would change the operator and is not done.

This is a family of scalar spectral assertions in a noncommutative algebra. It does not define a joint classical spectral measure for all $W_{f_i, \ell}$, and makes no estimate uniform in a growing tuple or a growing conductor.

9. TAPERS WITH EQUAL VARIANCE AND DIFFERENT WEYL GROWTH

The constant $1/12$ reflects where the variance lies among the translation lengths, not just its total size. We make this distinction quantitative for an explicit deformation of the actual arithmetic operator.

A boundary-profile criterion. Let $w \geq 0$ be continuous on $[0, 1]$, and set

$$J_w(u) = \int_0^1 w(a) |c_u(a)|^2 \, da, \quad K(a) = w(a)(1 - a).$$

Assume $K(0) = K(1) = 0$, $K(a) = K(1 - a)$, and that the periodic Fourier series of K is absolutely convergent with $\widehat{K}(n) \leq 0$ for every $n \neq 0$. The coefficients are real by symmetry. The overlap Cauchy–Schwarz argument and Fourier expansion give, for unit u ,

$$\begin{aligned} J_w(u) &\leq \frac{1}{2} \int_0^1 \int_0^1 K(x - y) |u(x)|^2 |u(y)|^2 dx dy \\ &= \frac{\widehat{K}(0)}{2} + \sum_{n \geq 1} \widehat{K}(n) |\widehat{F}(n)|^2 \leq \frac{\widehat{K}(0)}{2}. \end{aligned} \quad (9.1)$$

The symmetry of K implies

$$c(w) := \frac{\widehat{K}(0)}{2} = \int_0^1 w(a)(1 - a)^2 da = J_w(1), \quad (9.2)$$

so the bound is sharp. The same Tonelli argument as before handles every L^2 profile. More precisely the difference $c(w) - J_w(u)$ is the sum of $\sum_{n \geq 1} (-\widehat{K}(n)) |\widehat{F}(n)|^2$ and the overlap-variance term in (4.1) with a replaced by $w(a)$. The use of nonpositive nonconstant Fourier coefficients is the classical energy principle for a kernel of conditionally negative type; see [2]. The preceding calculation states the precise version needed here and proves it directly.

The symmetry condition is not arbitrary. For a bounded real mean-zero v , differentiate at $\eta = 0$ the profile $(1 + \eta v) / \|1 + \eta v\|_2$. The first variation is

$$2 \int_0^1 v(x) \left(\int_0^x K(a) da + \int_0^{1-x} K(a) da \right) dx.$$

It vanishes for every such v exactly when the bracket is constant. Its derivative is $K(x) - K(1 - x)$, so flat stationarity is equivalent to kernel symmetry. Imaginary first variations vanish automatically. The Fourier-sign condition is a sufficient global-optimality condition beyond stationarity; we do not assert that it is necessary.

There is also a finite-grid estimate without an asymptotic optimizer argument. For a unit vector in $\mathbb{C}^d$, set $F_i = |v_i|^2$ as before. Then

$$\begin{aligned} \frac{1}{d} \sum_{j=1}^{d-1} w(j/d) |c_j(v)|^2 &\leq \frac{1}{2} \sum_{i,k=0}^{d-1} K((i - k)/d) F_i F_k \\ &= c(w) + \frac{1}{2} \sum_{n \neq 0} \widehat{K}(n) \left| \sum_i F_i e^{2\pi i n i/d} \right|^2 \leq c(w). \end{aligned} \quad (9.3)$$

The kernel is periodic. All nonzero Fourier indices, including multiples of d , are retained; their signs help the bound. If w is Lipschitz and $d = q + 2M$, changing j/d to j/q and then enlarging the nonnegative sum proves

$$\frac{1}{q} \sum_{j=1}^{q-1} w(j/q) |c_j(v)|^2 \leq \frac{d}{q} c(w) + \text{Lip}(w) \frac{d}{q} \left(\frac{d}{q} - 1 \right). \quad (9.4)$$

Here $\sum_j |c_j(v)|^2 \leq d$ suffices for the error.

The full arithmetic deformation. Let $g \in C^1([0, 1]; \mathbb{R})$ be fixed, and suppose the profile $w(a) = ag(a)^2$ satisfies the criterion above with $c(w) > 0$. Absolute Fourier convergence is automatic in this class: the continuous periodic kernel has a weak derivative in L^2 , so Cauchy–Schwarz applied to its weighted Fourier coefficients gives $\sum_n |\widehat{K}(n)| < \infty$. Define

$$W_{f,\ell}^g = \sum_{p^m < e^\ell} b_f(p^m) g(m \log p / \ell) (S_{m \log p} + S_{-m \log p}). \quad (9.5)$$

For fixed f, g, C and $2 \leq |t| \leq C\ell$, the complete uniform conclusion is

$$c(w)t^2 - O_{f,g,C}(1) \leq \log \tau_\ell(e^{tW_{f,\ell}^g}) \leq c(w)t^2 + O_{f,g,C}(1 + |t|\sqrt{\log(2 + |t|)}). \quad (9.6)$$

We detail the changes to the pressure proof to keep every cost explicit. The local absolute and squared budgets are multiplied by at most $\|g\|_\infty$ and $\|g\|_\infty^2$, respectively. The relative Taylor argument therefore still restores all local harmonics at bounded cost for bounded physical coupling; each prime retains its single phase. The lower Bessel bound uses $F(a) = g(a)^2(1-a)^2$. Stieltjes integration by parts against (3.2), using $F(1) = 0$, gives

$$\sum_p b_p^2 F(h_p/\ell) = \ell^2 \int_0^1 a F(a) da + O_{f,g}(1) = c(w)\ell^2 + O_{f,g}(1).$$

The fourth-power remainder remains summable.

For the upper bound take $M = \lceil D_g t^2 \rceil$ and $q = \lceil t^6 \rceil$, choosing D_g large enough to absorb the factor $\|g\|_\infty^2$ in the unmasked word-tail estimate. The mask domination and relative Gaussian replacement are unchanged. In the normalized weak-variance sum, replacing $g(h_p/\ell)^2$ by $g(j_p/q)^2$ costs at most

$$\frac{\text{Lip}(g^2)}{q} \frac{V_f(\ell)}{\ell^2} = O_{f,g}(q^{-1}).$$

In each long bin the main mass is now $w(j/q)/q + g(j/q)^2/(2q^2)$; the summed second term is $O_g(d/q^2)$. The short-bin budget remains $O_{f,g}(\log^4(2+\ell)/\ell^2 + q^{-2})$. The moving-bin errors are unchanged up to a bounded factor. Equation (9.4) therefore bounds the weak-variance half by

$$c(w) + O_{f,g,C} \left(M/q + q^{-1} + \log^4(2+\ell)/\ell^2 \right).$$

The strong variance stays bounded. Gaussian concentration, the same uniform absorption of the arithmetic error, and restoration of the local strings prove (9.6). The constants are uniform on any C^1 -bounded family of tapers satisfying the stated kernel conditions.

The proof of Section 6 now gives rate $x^2/(4c(w))$ for every diverging scale $r_\ell = O(\ell)$ of $W_{f,\ell}^g(\ell r_\ell)$, with speed r_ℓ^2 . This uses $c(w) > 0$. It is not a central-law assertion obtained by sending r_ℓ to a constant.

An explicit equal-variance family. Take $g_\varepsilon(a) = \sqrt{1 + \varepsilon \cos(2\pi a)}$. The kernel is $K_\varepsilon(a) = a(1-a)(1 + \varepsilon \cos(2\pi a))$. Multiplying its absolutely convergent Bernoulli series gives

$$\begin{aligned} \widehat{K}_\varepsilon(0) &= \frac{1}{6} - \frac{\varepsilon}{2\pi^2}, & \widehat{K}_\varepsilon(1) &= -\frac{1}{2\pi^2} + \varepsilon \left(\frac{1}{12} - \frac{1}{16\pi^2} \right), \\ \widehat{K}_\varepsilon(n) &= -\frac{1}{2\pi^2 n^2} - \frac{\varepsilon}{4\pi^2} \left(\frac{1}{(n-1)^2} + \frac{1}{(n+1)^2} \right), & n &\geq 2. \end{aligned} \quad (9.7)$$

The $n=1$ restriction is $\varepsilon \leq 24/(4\pi^2 - 3)$. For $n \geq 2$ it is $\varepsilon \geq -(n^2 - 1)^2/(n^2(n^2 + 1))$. The positive ratio $(n^2 - 1)^2/[n^2(n^2 + 1)]$ increases with $n \geq 2$ (as a function of $y = n^2$ its derivative has numerator $(3y + 1)(y - 1)$). Thus all nonconstant coefficients are nonpositive exactly on

$$-\frac{9}{20} \leq \varepsilon \leq \frac{24}{4\pi^2 - 3}, \quad c_\varepsilon = \frac{1}{12} - \frac{\varepsilon}{4\pi^2} > 0. \quad (9.8)$$

This closed interval lies strictly inside $(-1, 1)$, so these tapers are uniformly smooth and bounded away from zero. The pressure estimate (9.6) holds uniformly over the entire interval, including its endpoints. This is the exact interval for the Fourier-sign certificate, not a claim that the flat profile ceases to be optimal outside it.

On a compact subinterval of its interior, both $w_\varepsilon(a)/a$ and $2\pi^2 n^2 [-\widehat{K}_\varepsilon(n)]$ have uniform positive lower bounds. The exact weighted deficit consequently dominates a positive constant times $D(u)$. Strong stability modulo affine phase follows from (1.11), uniformly on each such compact parameter set. No endpoint stability constant is asserted here.

This family has equal total leading first-prime variance:

$$\int_0^1 a g_\varepsilon(a)^2 da = \frac{1}{2}, \quad \sum_p b_p^2 g_\varepsilon(h_p/\ell)^2 = \frac{\ell^2}{2} + O_f(1),$$

uniformly over (9.8). The latter follows by the same bounded-variation integration against (3.2). Thus the unmasked product still has fixed-coupling logarithm $s^2\ell^2/2 + O_{f,s}(1)$, whereas its killed logarithm has leading coefficient s^2c_ε . The paired-word identity gives

$$\log \mathbb{E}F_\ell = -s^2 \left(\frac{5}{12} + \frac{\varepsilon}{4\pi^2} \right) \ell^2 + O_{f,s}(\ell\sqrt{\log(2+\ell)}). \quad (9.9)$$

The distribution of the paired words here uses the tapered first-prime coefficients. Formula (9.9) isolates a change in boundary confinement while preserving the total leading scalar variance.

Finally replace W by W^{g_ε} in (2.5), retaining the same gamma operator. Its fixed-window Weyl coefficient satisfies

$$\log K_{f,\ell}^\varepsilon = \left(\frac{1}{48} - \frac{\varepsilon}{16\pi^2} \right) \ell^2 + \log \frac{2\ell}{\sqrt{N}} + O_f(\ell\sqrt{\log(2+\ell)}), \quad (9.10)$$

again with bounded lower error and constants uniform in the displayed parameter interval. These are deformed logarithmic operators. Only $\varepsilon = 0$ is the original arithmetic operator of the preceding sections.

NOTE ON AI USE

The author declares the use of consumer-grade Gemini-3 and GPT-6 models for prior-art search, exploring and developing ideas and proofs, drafting and LaTeX typesetting; the author takes full and sole responsibility for all contents of this work.

REFERENCES

- [1] A. O. L. Atkin and J. Lehner, *Hecke operators on $\Gamma_0(m)$* , Math. Ann. **185** (1970), 134–160. doi:10.1007/BF01359701.
- [2] C. Berg, J. P. R. Christensen, and P. Ressel, *Harmonic Analysis on Semigroups: Theory of Positive Definite and Related Functions*, Graduate Texts in Mathematics, vol. 100, Springer, New York, 1984. doi:10.1007/978-1-4612-1128-0.
- [3] A. Bose, R. S. Hazra, and K. Saha, *Convergence of joint moments for independent random patterned matrices*, Ann. Probab. **39** (2011), no. 4, 1607–1620. doi:10.1214/10-AOP597; arXiv:1211.3864.
- [4] A. Bose, K. Saha, and P. Sen, *Some patterned matrices with independent entries*, Random Matrices Theory Appl. **10** (2021), article 2150030. doi:10.1142/S2010326321500301. Updated author version: arXiv:2203.05837v1 (2022).
- [5] W. Bryc, A. Dembo, and T. Jiang, *Spectral measure of large random Hankel, Markov and Toeplitz matrices*, Ann. Probab. **34** (2006), no. 1, 1–38. doi:10.1214/009117905000000495.
- [6] R. Arun Chandru, *Arithmetic identification from interval Weil spectra*, preprint, 2026. doi:10.5281/zenodo.22834353.
- [7] R. Arun Chandru, *Sharp Weyl laws and resonance transitions for logarithmic operators with killed translations*, preprint, 2026. doi:10.5281/zenodo.22881457.
- [8] M. Christ, *A sharpened Hausdorff–Young inequality*, preprint, 2014. arXiv:1406.1210.
- [9] P. Deligne, *La conjecture de Weil. I*, Publ. Math. Inst. Hautes Études Sci. **43** (1974), 273–307. doi:10.1007/BF02684373.
- [10] A. M. Garsia, E. Rodemich, and H. Rumsey, Jr., *On some extremal positive definite functions*, J. Math. Mech. **18** (1969), no. 9, 805–834.
- [11] T. Gillespie and G. Ji, *A prime number theorem for Rankin–Selberg L -functions over number fields*, preprint, 2009. arXiv:0910.3660.
- [12] L. Gross, *Logarithmic Sobolev inequalities*, Amer. J. Math. **97** (1975), no. 4, 1061–1083. doi:10.2307/2373688.
- [13] A. Laptev and T. Weth, *Spectral properties of the logarithmic Laplacian*, Anal. Math. Phys. **11** (2021), article 133. doi:10.1007/s13324-021-00527-y; arXiv:2009.03395.
- [14] M. Ledoux, *The Concentration of Measure Phenomenon*, Mathematical Surveys and Monographs, vol. 89, American Mathematical Society, Providence, RI, 2001.
- [15] A. D. Manov, *An extremal problem for positive definite functions with support in a ball*, Sb. Math. **215** (2024), no. 7, 920–931. doi:10.4213/sm10006e.
- [16] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, et al., eds., *NIST Digital Library of Mathematical Functions*, Section 5.11, asymptotic expansions of the gamma and digamma functions. <https://dlmf.nist.gov/5.11>.
- [17] A. Onatski, *Precise asymptotics for the norm of large random rectangular Toeplitz matrices*, preprint, 2025. arXiv:2509.04349.

- [18] A. Sen and B. Virág, *The top eigenvalue of the random Toeplitz matrix and the sine kernel*, Ann. Probab. **41** (2013), no. 6, 4050–4079. doi:10.1214/13-AOP863; arXiv:1109.5494.
- [19] J. A. Tropp, *User-friendly tail bounds for sums of random matrices*, Found. Comput. Math. **12** (2012), no. 4, 389–434. doi:10.1007/s10208-011-9099-z.

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